Taehyeon Kwon

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Education

Ph.D. **Pennsylvania State University**, University Park, PA — Aug 2023 – Present
Industrial Engineering (GPA: 3.68/4.00)
- Concentration: Operations Research
- Advisor: Dr. Anirudh Subramanyam

B.S. **Seoul National University of Science and Technology**, Seoul, South Korea — Mar 2017 – Feb 2023
Industrial Engineering (IT Management)
- Dual Degree with Northumbria University, Newcastle, UK (First Class Honours)
- Advisor: Dr. Min Kyu Sim

Research Interests

Scalable algorithms integrating **mathematical optimization** and **machine learning** for large-scale optimization problems under uncertainty.

- **Methodologies**: Stochastic Programming, Neural Network-Constrained Optimization, Reinforcement Learning
- **Applications**: Complex System Control & Optimization, Capacity Expansion Planning, Energy Storage Operation

Research Experience

Pennsylvania State University, Graduate Research Assistant — Aug 2023 – Present
University Park, PA
- Develop scalable ML-integrated optimization algorithms for capacity expansion planning.
- Experience with decomposition methods (progressive hedging, Benders) on HPC systems.
- Supported by U.S. Department of Energy (Award No. DE-SC0023361)

Seoul National University of Science and Technology, Undergraduate Research Assistant — Mar 2021 – Feb 2023
Seoul, South Korea
- Developed reinforcement learning frameworks for energy storage and microgrid control.
- Supported by National Research Foundation of Korea (NRF-2020R1F1A107387912)

Publications

[1] Machine Learning-Enabled Large-Scale Capacity Expansion Planning under Uncertainty.
Kwon, T., Subramanyam, A.
Electric Power Systems Research, 263, 113791 (2027)

[2] Optimal Microgrid Operation Considering Accurate Battery Degradation - World Model-based Reinforcement Learning Approach.
Park, J., **Kwon, T.**, and Sim, M.
Journal of Energy Storage, 141, 119151 (2026)

[3] A quadratic formulation of ESS degradation and optimal DC microgrid operation strategy using quadratic programming.
Choi, K. J., Park, J., **Kwon, T.**, Kwon, S., Kwon, D., Lee, Y. I., & Sim, M. K.
IEEE Access, 12, 88534-88546. (2024)

[4] Optimal Energy Storage System Control Framework Using Accurate Degradation Model – Reinforcement Learning Approach.
Park, J., **Kwon, T.**, and Sim, M.
Journal of Energy Storage, 71, 107964. (2023)

[5] Multi-Agent Reinforcement Learning Based Optimal PV-ESS Control In Grid.
Park, J.*, **Kwon, T.***, Kim, B.*, Hwang, Y., & Sim, M. K.
2023 IEEE IAS Global Conference on Renewable Energy and Hydrogen Technologies, (pp. 1-6). IEEE. (2023)
(equal contribution)*

[6] Optimal ESS Operation Based on Reinforcement Learning.
Kwon, T., Park, J., and Sim, M.
Proceedings of the Korea Information Science Society Conferences, 684-686. (2022)

Presentations

- Power Systems Computation Conference (PSCC), Limassol, Cyprus, Jun 2026, “Machine Learning-Enabled Large-Scale Capacity Expansion Planning under Uncertainty" (accepted).
- Institute for Computational and Data Sciences Symposium, University Park, PA, USA, Sep 2025, “Machine Learning-Enabled Large-Scale Transmission Capacity Expansion Planning".
- INFORMS Computing Society Conference, Toronto, ON, Canada, Mar 2025, “Machine Learning-Enabled Stochastic Capacity Expansion at Scale".

Teaching Experience

Teaching Assistant, IE 434 *Statistical Quality Control* — Spring 2025
Pennsylvania State University
Teaching Assistant, IE 425 *Simulation Modeling for Decision Support* — Spring 2024
Pennsylvania State University
Teaching Assistant, IISE 529 *Stochastic Processes* — Spring 2023
Seoul National University of Science and Technology

Honors & Awards

Max and Joan Schlienger Graduate Fellowship in Engineering — 2023-2024
College of Engineering, Pennsylvania State University

Glenn E. Singley Memorial Graduate Fellowship — 2023-2024
College of Engineering, Pennsylvania State University

FEGR Scholarship — 2023
College of Engineering, Pennsylvania State University

3rd Place, Artificial Intelligence Competition for Sustainable Energy — 2022
Korea Electronics Technology Institute

1st Place, Big Data Competition — 2021
Seoul National University of Science and Technology

Technical Skills

Operations Research: Stochastic Programming, Mixed-Integer Linear Programming, Linear Programming, Integer Programming, Nonlinear Programming
Machine Learning: Large-scale ML implementation, hyperparameter optimization, model validation
Programming Languages: Python, Julia
Optimization Tools: JuMP, Pyomo, Gurobi, CPLEX, HiGHS, IPOPT
HPC Systems: SLURM, Bash, Linux, high-performance parallel computing
Libraries: mpi-sppy, MPI, MPI4py, OMLT, MathOPTAI, PyTorch, Scikit-Learn, pandas, NumPy